AI101 MINI-PROJECT / TEAM 33

Predicting electrical output from a combined-cycle power plant

A linear regression study using plant operating conditions

Execution Lead
Trisha Bansal

Video Lead
Arinjay Garg

Documentation Lead
Shubham Maurya

PPT Lead
Jyoti

REPOSITORY
github.com/ArinjayGarg/Combined-Cycle-Power-Plant

DATASET & OBJECTIVE

Problem & Data

Estimate net hourly electrical output from ambient and exhaust conditions measured at a combined-cycle plant.

UCI Combined Cycle Power Plant | **9,568 Records** | **4 Predictors + 1 Target**

PREDICTORS (OPERATING VARIABLES)

- **AT** Ambient temperature (°C)
- **V** Exhaust vacuum (cm Hg)
- **AP** Ambient pressure (mbar)
- **RH** Relative humidity (%)

→

TARGET VARIABLE

PE

Net electrical output (MW)

Predicted from the four ambient & exhaust operating variables.

NOTEBOOK CHECK 0 missing values • 41 duplicate rows remained in the 9,568-row dataset.

Source: UCI Machine Learning Repository · Combined Cycle Power Plant (2006–2011)

METHODOLOGY & WORKFLOW

Approach

01

Inspect

Load 9,568 rows
Check types, nulls and duplicates

02

Explore

Review distributions, outliers and feature relationships

03

Split

80 / 20 train–test with random_state 42

04

Compare

Fit Linear Regression for 15 feature combinations

05

Evaluate

MAE, RMSE and R^2 plus 5-fold cross validation

LIBRARIES

NumPy · pandas · scikit-learn · Matplotlib · Seaborn

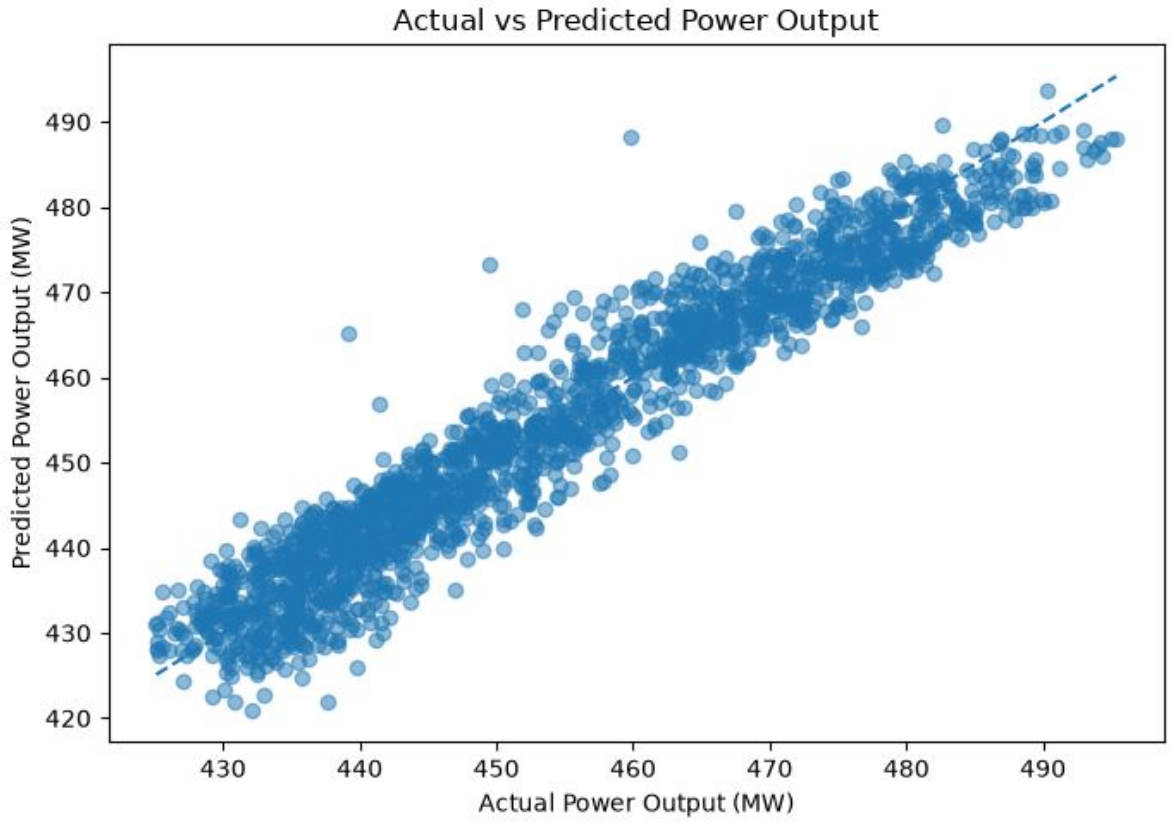
All four predictors were retained for the final fit after comparing every non-empty subset.

MODEL EVALUATION

Results

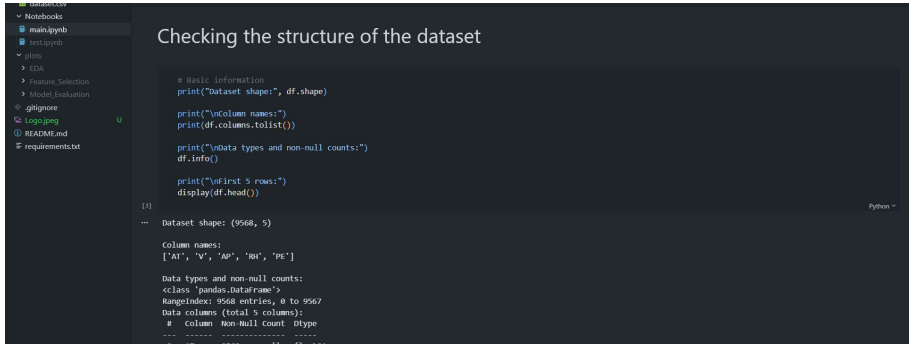
Best set: AT + V + AP + RH | **1,914 held-out records** | **15 feature sets compared**

ACTUAL VS PREDICTED POWER OUTPUT



Predictions track the observed output across the tested range.

PROJECT DEMO · notebook data inspection



TEST SET PERFORMANCE METRICS

R^2	RMSE	MAE
0.930	4.503 MW	3.596 MW

CROSS VALIDATION (5-FOLD)

R^2 0.928 · RMSE 4.572 MW

SUMMARY & REFLECTIONS

Learnings & contributions

CORRELATION ANALYSIS

r = −0.948

Ambient temperature has the strongest single-variable correlation with output.

MODEL SELECTION

4-Input Model #1

Ranks first overall; its RMSE gain over AT + V + RH is small (4.503 vs 4.513 MW).

VALIDATION CONSISTENCY

R^2 0.928 vs 0.930

Five-fold cross-validation results are extremely close to the held-out test set result.

TEAM CONTRIBUTIONS

Execution lead

Trisha Bansal

Idea selection and much of the coding

Video lead

Arinjay Garg

Team coordination and idea selection

Documentation lead

Shubham Maurya

Coding and documentation

PPT lead

Jyoti

Project ideas and documentation

AI-USE DISCLOSURE

AI assistance (Claude/ChatGPT) was used selectively for syntax debugging and initial slide layout drafts.